FINAL YEAR PROJECT · 2026 · AI / ML

Fake News Detection

A Leakage-Safe, Calibrated & ExplainableMLPipeline for Classifying News Authenticity

Kashish Behera
2301305076 · Model Development & Explainability

Dibyasanti Swain
2301305061 · Application Development & Integration

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INTRODUCTION

The Problem We're Solving

Classicaltextclassifierstrainedonpublicfake-newscorporaoftenlearn the wrong signal entirely —and once deployed, a staticlabelwith noconfidencescoreorexplanationishardfora readertotrust.



Publisher-Signature Leakage

Naive training on ISOT lets models key off "(Reuters)" datelines insteadof writingstyle —inflatingaccuracy past 99% on signal that vanishes in production.



No Confidence Calibration

ArawSVMmarginisn'taprobability.Withoutcalibration, a model can't tell a user how confident it actually is.



Black-Box Predictions

Abare"Real / Fake"labelgivesthereadernowayto see which words drove the decision, which erodes trust.



No Live Corroboration

Astaticclassifiercan'tcheckwhether a claim is actually being reported anywhere right now.

Existing Approaches & Their Gaps

Feature-by-feature,againsthowfake-newsclassifiersaretypicallybuiltandshipped.

Capability	Raw ISOT Baseline	Generic NB Classifier	Our System
Leakage-safe preprocessing	✗	✗	✓
Calibrated probability output	✗	✗	✓
Token-level explainability	✗	✗	✓
Live web corroboration	✗	✗	✓
Multi-modal input (URL / OCR / CSV)	✗	✗	✓

Note: baseline rows reflect typical published approaches on this task, not benchmark numbers we ran ourselves.

METHODOLOGY

System Architecture

Frontend Layer

- Streamlit Web App
- Manual Text Entry
- URL Scraper (BeautifulSoup)
- OCR Image Upload
- Batch CSV Processor

Model Layer

- TF-IDF Vectorizer
- Calibrated SVM (LinearSVC)
- Linear Explainability Engine
- Clickbait Heuristic Scorer

Data Layer

- SQLite —predictions log
- SQLite —live_corpus cache
- NewsAPI (live verification)

HOW DATA FLOWS Input → cleaned & vectorized → classified by the calibrated SVM → explained token-by-token → optionally cross-checked live via NewsAPI → logged.

OUR PRODUCT – USER & HOW IT WORKS

An AI-powered application that detects, explains and verifies news to help users make informed decisions.

WHO CAN USE IT?

Students & Researchers
Check news/articles for suspicious patterns

General Public
Analyze news before sharing it

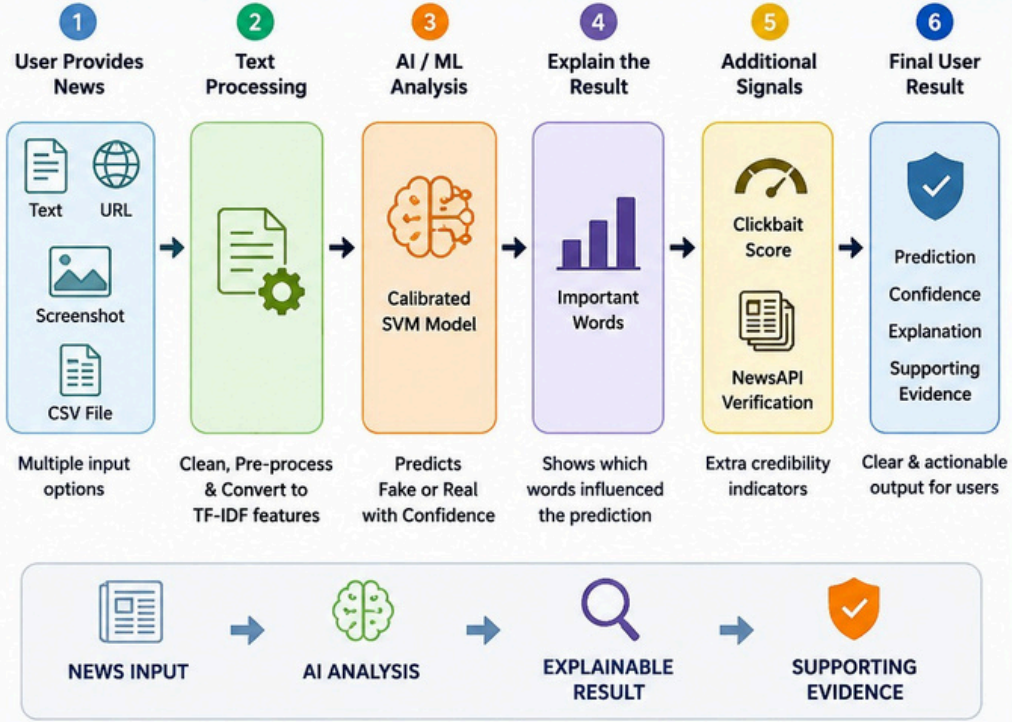
Journalists & Content Creators
Get an initial credibility signal

Educators
Demonstrate misinformation and AI concepts

Organizations
Use as a preliminary screening tool

This is a decision-support tool, not a replacement for professional fact-checking.

HOW DOES IT WORK?



Our system combines AI prediction, explainability and real-time verification to promote a more informed and responsible information ecosystem.

KEY FEATURE 01

Leakage-Safe Preprocessing

The Evidence

99.8% of real ISOT articles literally contain the word "Reuters." Only 1.4% of fake articles do. A model can hit near-perfect accuracy by detecting a publisher tag, not a writing style.

21,378 / 21,417

Real articles mentioning "Reuters" (99.8%)

322 / 23,481

Fakearticlesmentioning"Reuters" (1.4%)

Fix: regex-strip the "CITY (Reuters) -" dateline and any remaining "Reuters" mention from every title and body, before vectorizing.

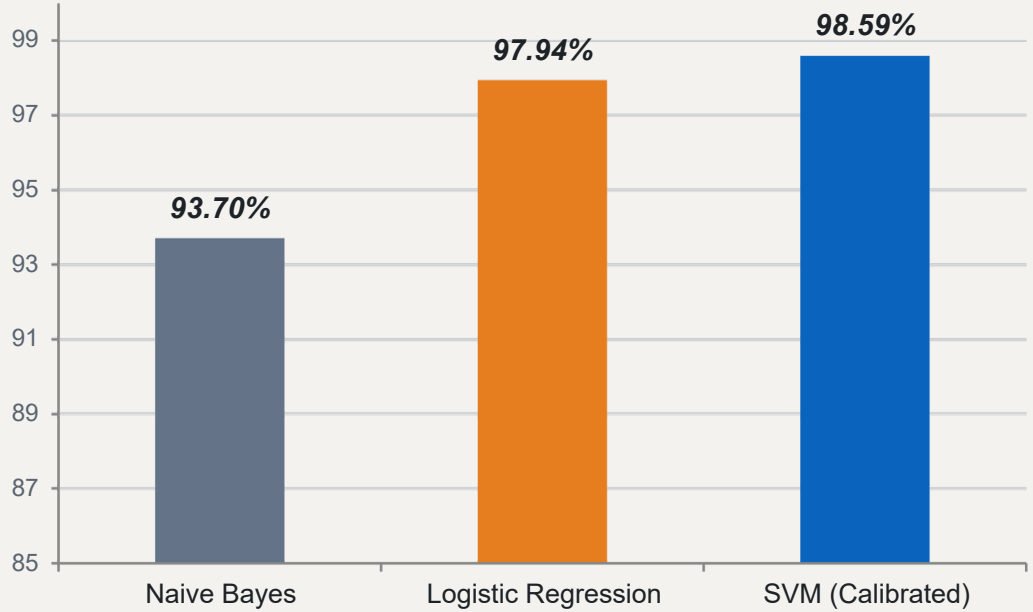
Dataset Pipeline (ISOT)

Raw articles (Fake + Real)	44,898
After Reuters/dateline strip	44,898
After title+text dedup	39,105
Train split (80%)	31,284
Test split (20%, held out)	7,821

KEY FEATURE 02

Model Comparison & Selection

Threeclassifierstrainedonidentical,leakage-safeTF-IDFfeatures,evaluatedon the same held-out 7,821-article test set. BestmodelselectedbyF1 score.



SVM (LinearSVC)

Selected —highest F1

Accuracy	**98.59%**
Precision	**98.38%**
Recall	**99.03%**
F1 Score	**98.71%**

KEY FEATURE 03

Calibrated SVM Core

LinearSVC has no predict_proba —it outputs a decision margin, not a probability. We wrap it in CalibratedClassifierCV (3-fold Platt scaling) so the app can show a genuine confidence score.

Confusion Matrix —Test Set (n = 7,821)

	Pred: Fake	Pred: Real
Actual: Fake	3,523	59
Actual: Real	56	4,183

Final Deployed Metrics

Accuracy	98.59%
Precision	98.38%
Recall	99.03%
F1 Score	98.71%
Calibration	Platt (CV=3)

KEY FEATURE 04

Explainability Engine

Foralinearmodel,SHAPvalue≈TF-IDFweight×learned coefficient, per token. We compute this directly —no black-box surrogate — and surface it two ways.

Token Highlighting

Renders the input article inline, color-coding each word red (pushes Fake) or green (pushes Real) by its signed contribution.

Linear Attribution Chart

Ranks the top 5 words pushing each direction as a horizontalbar chart—the same mathafullSHAP library would compute for a linear kernel.

Scope note: this is a manual linear-attribution implementation —it does not import the shap package. Mathematically equivalent for a linear model, worth stating precisely if asked.

KEY FEATURE 05

Live Verification via NewsAPI

Theclassifierisstyle-based—itcan'tknowifaclaimistrue.Thismoduleadds a complementary, independent signal: is anyone else reporting it right now?



API key configured

Live corroboration runs; similarity score against real-time search results is shown alongside the classifier's label.

No API key set

App falls back honestly: "Linguistic Style Check Only —short claims can't be verified without an API key."

KEY FEATURE 06

Multi-Modal Input

Fourindependentwaystogetanarticleintothe classifier —because real users rarely start with clean plain text.



Manual Text

Paste or type an article body di r ect l y.



URL Scraper

BeautifulSoup pulls title + paragraph text from any live news link.



OCR Upload

Screenshot upload → OCR.space → extracted text.



Batch CSV

Upload a CSV, pick a text column, classify every row, export results.

KEY FEATURE 07

Clickbait & Credibility Signals

Arule-basedlayer,fullyindependentoftheMLclassifier—deterministic,explainable on its own, and useful even when the modelis uncertain.

Clickbait Score Inputs

- Sensationaltriggerwords(“shocking”, “exposed”, “bombshell”…) — up to 45 pts
- ALL-CAPS word ratio —up to 25 pts
- Exclamation / question mark density —up to 20 pts
- Leading list-number pattern (“7 Reasons…”) —10 pts

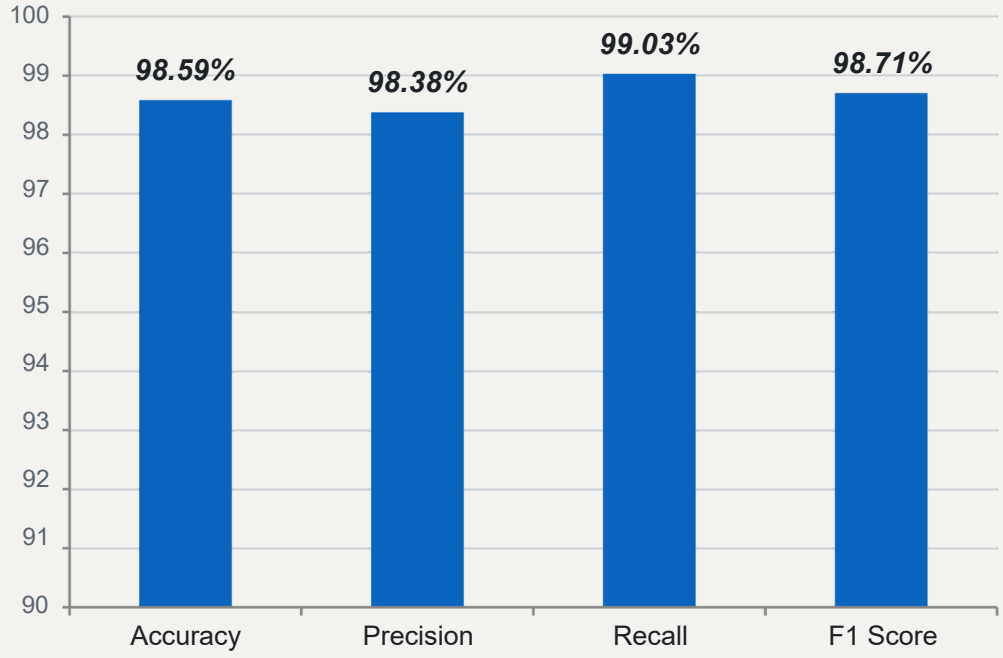
Design Note

This is deliberately not machine-learned. It runs alongside the SVM as a transparent, second opinion —a high clickbait score doesn't override the model's label, it's shown next to it so the reader can weigh both.

EVALUATION

Results & Performance

Everymodelevaluatedonthesameheld-out7,821-articletest set, never seen during training.



98.59%

Finalaccuracy—Calibrated SVM on unseen test data

+4.89 pts

Absoluteimprovementoverthe weakest baseline tested (Naive Bayes)

39,105

Deduplicated,leakage-safe articles used for training + evaluation

IMPLEMENTATION

Tech Stack

scikit-learn
TF-IDF, SVM, calibration

pandas
Dataset loading & cleaning

St ream lit
Web application UI

BeautifulSoup4
URL article scraping

requests
NewsAPI & OCR calls

matplotlib
Charts & calibration plot

w ordcloud
Word-frequency visuals

SQLite3
Prediction & session logs

NewsAPI
Live corroboration search

OCR.space
Screenshot text extraction

Python 3.12
Core language

CalibratedClassifierCV
Platt-scaled probabilities

WRAPPING UP

Conclusion & Limitations

What We Achieved

- Leakage-safe preprocessing, verified on 44,898 raw articles
- Calibrated SVM at 98.59% accuracy / 98.71% F1
- Manual linear explainability, no black box
- Optional live corroboration via NewsAPI
- Four input modes: text, URL, OCR, batch CSV

Current Limitations

- Session login is a username label, not real authentication
- OCR relies on a shared, rate-limited public demo key
- A style classifier is not a fact-checker —short, out-of-domain claims are a known hard case
- English-language articles only

WRAPPING UP

Future Scope

Concretenextsteps,notcompletedwork —each item below is a proposed extension, not a claimed result.

Multilingual Support

Extend beyond English to Hindi, Odia, and other regional news content.

Transformer Fine-Tuning

Evaluate a fine-tuned transformer (e.g. DistilBERT) head-to-head against the SVM on the same held-out split.

Real Authentication

Replace the username-only session with proper login and access control.

Cross-Domain Evaluation

Testgeneralizationonshort-claim,out-of-domain datasets such as LIAR, and report the honest accuracy drop.

Mobile / Browser Extension

Package the classifier for real-time, on-the-spot use outside the dashboard.

Multimedia Verification

Extend detection to image and video-based mis inf or m at i on.

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Thank You

Kashish Behera | Dibyasanti Swain